Case Study – Risks Analysis and Security Policies for a LoRa Underground Mine Network

Pattarapol Tantechasa
Student
Swinburne University of Technology
Melbourne, VIC, Australia
103883220@student.swin.edu.au

***Executive Summary*— In underground mining environment, traditional communication infrastructure such as Wi-Fi and optical fiber is vulnerable to damage during emergencies, which could lead to failures in coordinating rescue operations. A LoRa-based linear sensor network has been proposed as a low-cost, rapidly deployable solution to enable communication in emergencies, supporting the transmission of location data for personnel and equipment. However, the system still has many vulnerabilities that could be improved. This report presents a structured risk analysis of the proposed system using the Delphi method, focusing on five key threats: message loss due to flooding protocol, lack of encryption and authentication, physical tampering of relay nodes, relay node failure, and network flooding from malfunctioning tags. Security policies were formulated to address the highest-priority risks, followed by practical implementation for each policy, summaries and recommendations. The proposed security enhancements increase the resilience and trustworthiness of LoRa-Based communication systems in high-risk underground environments.**

Keywords*— *LoRa-Based, Network, Underground, Risk Analysis, Delphi Method, Relay Nodes, Tags, Tampering, Symmetric Cryptography, Flooding, Forwarding.

I. Introduction

In underground mining environments, using traditional communication infrastructure such as Wi-Fi and optic fiber could be damaged during emergencies, which creates vulnerabilities in the system. The system needs new solution to be able to maintain communication with personnel involved in rescue or recover tasks. LoRa(Long Range)-Based Linear Sensor Network has been proposed, it transmitting location information of personnel and equipment in an underground mine. Tags carried by personnel transmit location data, which is forwarded through relays using a simple flooding protocol until raching a headend.

While the system is practical and cost-effective, and easy to deploy, it lacks critical security features. Message could be lost due to flooding protocol and could be suffered from unfairness. Moreover, without encryption or authentication, sensitive location data may be culnerable to interception or tampering. The system needs a well-defined security plan to ensure the system's reliability, integrity, and confidentiality. This report aims to presents a structured analysis of the system's vulnerabilities, prioritises risks using Delphi method, and proposes both security policies and a practical implementation plan.

II. Risk Analysis

This section identifies and evaluates key risks and vulnerabilities associated with the LoRa-based emergency communications system for underground mining environment. To systematically assess these risks, the Delphi Method was applied, which scores each risk based on two factors: Likelihood (L) and Impact (I). Each risk is then prioritized based on a combined risk score (L × I).

TABLE I. Risk Analysis with Delphi Method

ID	Risk Description	L	I	Score	Priority
R1	Message loss due to flooding protocol	5	5	25	High
R2	Lack of encryption or authentication	2	5	10	Low
R3	Physical tampering of relays nodes	3	5	15	Medium
R4	Relay node failure	3	3	9	Low
R5	Network flooding or jamming by bad node	2	5	10	Low

A. *Risk 1: Message loss due to flooding protocol*

The system uses a flooding-based message forwarding protocol, where relay nodes receive a message and rebroadcasts it. This makes the system simple to set up and operate, it also introduces a high risk of message collisions, duplication and loss, which creates vulnerabilities to the relay hardware. This is more likely to occur when multiple tags are active at once, or when the number of relay nodes increases. As well as when tags are located further from the headend, messages must traverse more hops, which increases the chance message loss. Linear network structure makes the risk much more critical because when failure occurs at one point, it affects all nodes downstream. This directly affects the impact of system reliability, especially during emergency situations. Therefore, this risk has been assigned a Likelihood of 5 due to the system design, and Impact of 5 because message loss could endanger lives.

B. *Risk 2: Lack of encryption and authentication*

The system lacks encryption and authentication, making the system hardware like relay nodes and the headend vulnerable to eavesdropping and spoofing. Location data could be leaked without encryption and attackers could infect false messages without authentication. This creates huge vulnerability and serious risks in rescue operations where accurate location data is vital. An attacker could confuse coordination efforts or compromise safety by faking tag data. However, this requires attackers have a good technical setup and proximity. This risk has been assigned to the Likelihood of 2 since it is very rare for attacks to happen where the system is deployed two kilometers underground. Another hand, the Impact is 5 because successful breach could mislead rescue teams or expose sensitive data.

C. *Risk 3: Physical Tampering of Relays Node*

There is minimal protection and no tamper detection for relay nodes placed, making them vulnerable to physical interference. There is a moderate chance that devices may be damaged by machinery, removed by mistake or intentionally tampered with in a mining environment. The network may become partially or fully disconnected beyond when a critical relay is taken out. The system has no mechanism to alert operators if a relay goes offline. Likelihood has been assigned to 3 due to the realistic chance of physical disturbance, and an Impact of 5 especially if relay near the headend was taken out, which could fail the whole communication in the linear network system.

D. *Risk 4: Relay Nodes Failure*

In underground environments, relay nodes are exposed to harsh conditions like dust, water, vibration, and extreme temperature. It also relies on batteries with limited lifespan. The system doesn't include any kind of health monitoring or status reporting for the relays. This results in undetected relay nodes failures until communication is lost, it could isolate multiple downstream tags if any relay nodes fail. Likelihood is rated as 3 since such failures are expected over time, and Impact as 3 because the disruption depends on where the failure occurs, some node failures may have minimal impact but failures near the headend could disrupt communication for the entire network.

E. *Risk 5: Network Flooding or Jamming by Bad Node*

The uses of flooding-based message forwarding protocol create vulnerabilities of disruption to the entire network due to an excessive amount of malfunctioning or malicious sent by tags. The system lacks rate-limiting or access control, meaning a single tag transmitting continuously could saturate the relays, which prevents other messages from being processed or forwarded. The Likelihood has been assigned as 2 due to small chances of an attack with this nature but is possible in case of firmware bugs. The Impact is rated as 5 because it could disrupt the entire network, making it useless during an emergency. Rescue teams would be unable to track personnel or receive updates from the affected zone.

III. Policy Formation

This section outlines specific security policies to mitigate the most critical threats of the proposed LoRa-based communication system presented in the previous section. These policies are designed to address issues related to data loss, lack of communication security, physical vulnerabilities, and potential abuse of the flooding protocol.

A. *Policy 1: Intelligent Message Management*

To ensure reliable delivery of critical location data, the system must enforce intelligent message management at each relay node, including mechanisms for identifying and discarding duplicate messages, implementing queue management, and prioritizing messages.

B. *Policy 2: Applies Symmetric Cryptography*

To protect the confidentiality and integrity of transmitted data, all communication between tags, relay nodes, and the headend must be encrypted and authenticated. This policy applies uniformly across all devices, including tags, relay node and the headend, and must be enforced before deployment. Keys must be securely stored on each device.

C. *Policy 3: Physical Protection of Relay Nodes*

Relay nodes must be physically secured against unauthorize access, environmental hazard and deliberate tampering. All relay nodes must be deployed with tamper-evident hardware like sealed enclosures. Relay nodes are vulnerable to accidental knocks from equipment, theft or interference from personnel.

D. *Policy 4: Rate Limiting and Anomal Detection*

To prevent disruption from malfunctioning or malicious tags, all relay nodes must enforce rate-limiting controls and monitor incoming message patterns for anomalies.

IV. Implementation of Security Policies

A. *Implementation for Policy 1: Intelligent Message Management*

To implement intelligent message management, each relay node should have a small buffer and message history table. Message should be assigned unique identifiers, such as tag ID plus timestamp or sequence number, so relays could recognize duplicates. Relay should check message ID against its recent history before deciding to forward message to other relays. This prevents message collision and duplication in communication of the system. The system must implement a queue management using first-in, first-out (FIFO) buffer with prioritization logic to ensure older or urgent messages are forwarded first. This helps avoid overloading the network and reduces unfair message loss for distant tags. Implementation can be done using lightweight logic on the relay's microcontroller, requiring minimal computational overhead. Forwarding policies can be tuned during field testing based on traffic patterns to find a balance between responsiveness and network load.

B. *Implementation for Policy 2: Applies Symmetric Cryptography*

All devices in the system, including tags, relay nodes and the headend, must be provisioned with a shared symmetric key during initial setup. The key is used to encrypt and decrypt messages using efficient algorithms such as AES-128 [3]. Each message should include a Message Authentication Code (MAC) to ensure message integrity. When a device like relay nodes and the headend receives a message, it decrypts the content and verifies the MAC before accepting or forwarding it. Keys must be stored securely on the device, such as protected flash memory. This computational lightweight implementation suits well with resource-constrained LoRa devices. During devices setup process, the key should be distributed to relay nodes and the headend.

C. *Implementation for Policy 3: Physical Protection of Relay Nodes*

To physically secure relay devices, all units should be housed in IP-rated enclosures, ideally IP65 or above [5], to provide dust and water resistance. Breakable security seals or magnetic switches should be included for tamper-evident designs. Log should be generated for any tamper event detected and should send a "tamper alert" message to the headend. Brackets or anchors should be used during the installation of devices to ensure it securely in fixed positions withing the mine, this prevents movement, theft and intentional tampering. As well as personnel should follow a documented checklist that includes verifying enclosure integrity and testing alert mechanisms. To detect early signs of physical interference, an inspection routine should be conducted and ensure logs from relay nodes gets reviewed regularly.

D. *Implementation for Policy 4: Rate Limitting and Anomal Detection*

Each relay node must track incoming messages by sender ID and timestamp. The system must apply simple rate-limiting algorithm, for instance, no more than one message per tag every 10 seconds. The relay nodes either drop or flags the message from that tag if it exceeds the threshold. Message counters and timestamps should be maintained in short-term memory to allow real-time enforcement. Relays should store a basic log of sender activity, which includes IDs, message count and time window for further analysis. Logs can be downloaded periodically for inspection or in case of network incidents. These controls can be implemented using basic software routines on the relay's microcontroller, ensuring low overhead.

V. SUMMARY AND RECOMMENDATIONS

This report explored key vulnerabilities in the proposed LoRa-based emergency communications system for underground mines. The analysis with Delphi method identified critical risks including message loss from the flooding protocol, lack of encryption and authentication, physical tampering of relay nodes, and potential flooding or jamming from malfunctioning tags. Furthermore, security policies and implementation were proposed to each risk, strategies were outlined using lightweight technologies suitable for the resource-constrained environment.

A. *Recommendations*

1) Duplicate Filtering and Prioritised Message Queuing

Relay nodes should use message ID and timestamp to detect and filter out duplicate messages. The system must apply to a FIFO-based queue with message prioritization to ensure critical or older messages are forwarded first. This will reduce congestion, minimize packet loss and improve fairness for distant tags in the flooding-based network.

2) AES-128 Encryption with Message Authentication

All messages exchanged between tags and relays, and the headend must be encrypted using AES-128. As well as including a Message Authentication Code (MAC) for data integrity. This will prevent eavesdropping and spoofing, protecting the confidentiality and integrity of location data within the network.

3) Physically Secure Relay Nodes with Tampering Detection

Relay nodes should be enclosed in IP-rated, tamper-evident hardware and securely mounted in fixed positions. The headend should be alerted when any tamper event occurs to any device, and all tamper events should be logged. Regular inspection and maintenance routines must be established to detect and respond to physical interference or damage.

4) Rate Limiting and Detect Abnormal Message Patterns

Each relay node must monitor message frequency per tag and enforce rate-limiting rules. If a tag exceeds its threshold, the relay should drop or flag the messages. This prevents network flooding from faulty or malicious nodes and ensures consistent communication availability.

5) Securely Provision and Manage Shared Symmetric Keys

Each device in the network should securely store symmetric encryption key during setup, it should be stored in protected memory, and future support for key updates or rotation. This enhances the system's long-term security.

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